MIL-G-19287C(SHIPS)
25 June 1969
SUPERSEDING
MIL-G-19287B(SHIPS)
20 April 1961
(See 6.4)

MILITARY SPECIFICATION

GENERATOR SET, STEAM TURBINE,

2500 KW, 450 VOLT, 60 CYCLE, A.C.

1. SCOPE

1.1 Scope. This specification covers steam turbine generator sets for Naval shipboard use.

1.2 Classification. The turbine generator sets shall be of the following types, as specified (see 6.1):

Type I - Condensing, constant speed operation.
Type II - Noncondensing, variable speed operation.
Type III - Condensing, constant and variable speed operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-691 - Steel Plate, Carbon-Silicon, Carbon Molybdenum and Manganese-Molybdenum Alloys Hot Rolled (Marine Boiler Quality).

MILITARY

MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-R-2729 - Regulator Set, Voltage A.C. Generator (Naval Shipboard Use).
MIL-G-3087 - Generator Sets, Steam Turbine (Direct and Alternating Current) (Naval Shipboard Use).
MIL-G-3124 - Generator, Alternating Current, 60 Cycle (Naval Shipboard Use).
MIL-Q-9858 - Quality Program Requirements.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shiboard Use).
MIL-C-15430 - Condenser, Steam, Surface; (Naval Shipboard).
MIL-E-15465 - Ejector Assemblies, Air.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-S-16113 - Steel Plate, High Tensile (HT), Hull and Structural.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use).
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of.
MIL-T-17523 - Turbine, Steam, General Auxiliary (Naval Shipboard Use).
MIL-E-17555 - Electronic And Electrical Equipment and Associated Repair Parts, Preparation For Delivery of.
MIL-C-19836 - Coolers, Fluid, Industrial, Air Motor and Generator, Naval Shipboard.
MIL-I-24092 - Insulating Varnish, Electrical Impregnating.

STANDARDS

MILITARY

MIL-STD-167 - Vibration Requirements for Shipboard Equipment.

FSC 6115

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DRAWINGS

NAVAL SHIP SYSTEMS COMMAND
9000-S6201-74453 - Navy Standard Switchboard, Units, Type and Overall Dimensions of Units.
5000-S4824-1385797 - Valves, Steel Flanged, Globe, Combined Exhaust and Relief, 50 PSI Maximum at 775°F.
5000-S4824-1385798 - Valves, Steel Flanged, Angle Combined Exhaust and Relief, 50 PSI Maximum at 775°F.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Complete turbine generator sets.

3.1.1 Type I, condensing, constant speed operation. The turbine generator sets shall be in accordance with MIL-G-3087, except as otherwise specified herein.

3.1.1.1 The turbine generator sets shall have the following classification:

Type required	Salient pole.
Steam pressure (at throttle)	As specified (see 6.1)
Steam temperature (at throttle)	Dry-saturated (1.0 percent maximum moisture).
Exhaust pressure	As specified (see 6.1)
Parallel operation requirements (see 3.1.1.9)	Parallel operation required.
Rating:	
Kilowatt	2500 (maximum rating).
Voltage	450 volts alternating current (a.c.)
Class of insulation	Class B or H.
Generator speed	1200 revolutions per minute (r.p.m.).
Degree of enclosure	Water-air-cooled.
Type of voltage regulator	Rotary amplifier or static (types II or III of MIL-R-2729).
Regulators per switchboard	One main.
Repair parts	As specified herein.
Frequency	60 cycles per second (c.p.s.).

3.1.1.2 Each turbine generator set shall be comprised of the turbine, generator, reduction gear, subbase, and oil coolers mounted as an integral unit, and a condenser and air ejector of the separately mounted type. Integral hand lubricating oil pumps for starting shall be included.

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3.1.1.3 The generator set, including all attached accessories, shall be as lightweight and compact as possible consistent with design and performance requirements specified herein. If necessary, weight or size restrictions shall be as specified (see 6.1).

3.1.1.4 At rated load condition, each complete generator set shall be capable of withstanding, for 10 seconds, without damage to any parts a three-phase, fully asymmetrical short circuit at the switchboard (assuming 30 feet of interconnecting cable from generator terminals to switchboard). The contractor shall submit a summation of calculations on the anticipated short circuit torque at the generator air gaps, the elastic characteristics of all rotors, the shaft and gear loading characteristic and the short circuit torque multiplying factor. This data shall be accompanied by the manufacturer's statement that all factors have been considered in designing for this short circuit torque condition and that all parts will withstand this loading without damage. Data shall be submitted with the outline (installation) drawing for review.

3.1.1.5 Shock. The complete turbine generator set, including attached and unattached accessories, shall conform to the requirements of the Appendix.

3.1.1.6 Inclination. The equipment shall be designed to operate satisfactorily in accordance with this specification under the following conditions:

(a) When the ship is permanently trimmed down by bow or stern as much as 5 degrees from the normal horizontal plane.
(b) When the ship is permanently listed up to 15 degrees to either side of the vertical.
(c) When the ship is pitching 10 degrees up or down from its normal horizontal plane.
(d) When the ship is rolling up to 45 degrees to either side of the vertical.

3.1.1.7 Turbines, gears, and couplings.

3.1.1.7.1 The turbines, gears, and couplings shall be in accordance with MIL-T-17523, except as otherwise specified herein.

3.1.1.7.2 The "emergency power" requirements shall not apply.

3.1.1.7.3 The governor synchronizing motor shall be a single phase, readily reversible, commutatorless, a.c. motor in accordance with MIL-M-17059.

3.1.1.7.4 The maximum acceptable reading (double amplitude) of the vibration indicator shall not exceed 0.001 inch as measured on the bearing cap or 0.002 inch as measured on the rotating shaft adjacent to the bearings. These limits shall not be exceeded at any speed up to 25 percent above designed operating speed.

3.1.1.8 Lube-oil coolers. The turbine generator lubricating oil coolers shall be designed and constructed in accordance with MIL-C-15730 and shall form an integral part of the turbine generator unit.

3.1.1.8.1 The oil coolers shall be type A, class 2 of MIL-C-15730 for 95°F salt water cooling of the turbine generator lubricating oil.

3.1.1.8.2 The cooler tubes may be of any of the sizes specified in MIL-C-15730.

3.1.1.9 The combined units shall be capable of satisfactory operation in parallel with steam at any pressure throughout the pressure range but with the maximum difference in pressure between units no greater than 20 percent. The overall speed drop of the units from 0 to 100 percent load may be 4 percent. With the maximum difference in pressure between units at 20 percent, the difference in load division between units shall not exceed 25 percent. With the pressure the same for both units, the units shall meet the load division requirements of MIL-G-3087. No more than two units will be required to operate in parallel at any one time.

3.1.1.10 Condensers. The condensers shall be designed and constructed in accordance with type II, class B of MIL-C-15430 except as modified by 3.1.1.10.1 and 3.1.1.10.2.

3.1.1.10.1 The condenser shall be designed to maintain pressure at the turbine exhaust flange as specified (see 6.1) when the turbine generator is operating at full rated power.

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3.1.1.10.2 The condenser shall be 2 pass with 8 foot tube length, and shall be equipped with double tube sheets at each end. The material in the inner tube sheet shall be as specified for the outer tube sheet or shall be steel conforming to class A of QQ-S-691 or grade HT, type I of MIL-S-16113. A space of at least 1/2 inch between the tube sheets shall be provided by machining the face of one or both tube sheets. This space shall be drained to a reservoir provided with a salinity cell. The joint between the tube sheets shall be gasketed or welded. Outer tube sheets shall be drilled as specified in MIL-C-15430. The holes in the inner tube sheet shall be drilled 4-1/64 inches in diameter. The tubes shall be expanded in the outer tube sheet only. The condenser shell shall not exceed 3 feet 3 inches in diameter.

3.1.1.11 Air ejectors. The turbine generator condenser air ejector assemblies shall be designed and constructed in accordance with type A of MIL-E-15465.

3.1.1.11.1 The air ejector assemblies shall be designed for operation with steam conditions as specified (see 6.1).

3.1.1.11.2 Each air ejector shall have sufficient capacity expressed in pounds per hour free dry air plus vapor to saturate at 7.5°F. below vacuum temperature to serve the condenser at full load condenser vacuum and also at 1.5 inches Hg. absolute partial load condition.

3.1.1.11.3 The allowable pressure drop when circulating full load condensate shall be 5 p.s.i.

3.1.1.11.4 There are no specific limitation as to weight, space, foundations and sizes other than as specified in 3.1.1.10.2.

3.1.1.11.5 A gland exhaust condenser is not required.

3.1.1.11.6 Inter and after condensers shall be designed for 90 p.s.i. pressure from the condensate pumps.

3.1.1.12 Generator and excitation-regulation system.

3.1.1.12.1 Generators shall conform to MIL-G-3124 except as otherwise specified herein.

3.1.1.12.1.1 When class H insulation is used, the equipment shall conform to the insulation suitability requirements specified in MIL-E-917. Reference to the approval letter shall be shown on the master drawing (see 3.3).

3.1.1.12.1.2 Except for shafts and identification plates, exterior surfaces of the generator shall be painted in accordance with MIL-E-917. Electrical insulation, surfaces in contact with lubricating oil or grease, commutators, collector rings, brushes, bearings, and other rotating parts of the machine from which centrifugal force may cause the paint to be thrown on the winding when the machine is operating at rated load and speed and at normal temperature, shall not be painted; insulation varnish conforming to MIL-I-24092 may be applied to such parts. All other parts that are not corrosion-resistant (see MIL-E-917 for listing of corrosion-resistant materials and treatments) shall be painted in accordance with MIL-E-917 except that one coat of enamel of light color will be sufficient.

3.1.1.12.1.3 Generators shall be capable of withstanding without injury to any part, a three-phase short circuit at the terminals for a period of 30 seconds with constant field excitation corresponding to the value of ceiling excitation provided when the associated voltage regulator is in operation. The generator shall withstand, without injury, a single-phase short circuit between any two terminals for a period of 30 seconds with constant field excitation corresponding to the value of ceiling excitation provided when the associated voltage regulator is in operation.

3.1.1.12.1.4 Overload. At rated voltage and frequency and at a power factor of 0.50, each generator shall be capable of delivering 150 percent rated current for 1 minute.

3.1.1.12.1.5 The generator efficiency shall be not less than 95.0 percent at rated load.

3.1.1.12.1.6 The deviation factor of the open-circuit terminal voltage shall not exceed 6 percent on any or all phases.

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3.1.1.12.1.7 Harmonic content. From no load to full load, the root means square value of the sum of all of the harmonic parts in the output voltage wave shall not exceed 5 percent of the root means square value of the rated output voltage. The root mean square value of any single part shall not exceed 3 percent of the root mean square value of the rated output voltage. Harmonic of 0.2 percent or less of the root mean square value of the rated output voltage may be disregarded in the determination of the harmonic content of the output voltage wave.

3.1.1.12.1.8 The generator air cooler shall conform to the requirements of class 2 of MIL-C-19836.

3.1.1.12.2 When static (type III) excitation-regulation systems are used, the power rectifiers (and reactors if feasible) shall be mounted in the cooling air stream of the generators, and those rectifiers (and reactors) shall be readily accessible for inspection, removal, and disassembly or replacement. Load current transformers and associated excitation assemblies (not mounted in the generator cooler enclosure) shall be mounted in a dripproof enclosure arranged for deck mounting. Voltage regular assemblies, including signal current transformers, shall be designed and arranged for mounting in the associated switchboard (furnished by others) and shall, in general, fit into the space available for voltage regulator assemblies on the standard switchboard shown on Drawing 9000-S6201-74453.

3.1.1.12.3 Connections to the voltage regulator equipment shall be electrically isolated from the generator power leads by means of isolating transformers or other means approved by the procuring activity.

3.1.1.12.4 Voltage regulator manual and auto voltage adjusting rheostats shall be designed for manual control locally and for electic control from a remote station. The system shall be controllable by a triple pole, double throw switch having a spring return off position. The rheostat drive motors may be a reversible series motor with self-contained brake which will release when the motor is energized, and will be applied when the motor is deenergized. The gear and coupling shall prevent damage to the motor when the end stops are reached, and shall permit manual override. Switches are not required to be furnished under this specification.

3.1.1.13 The use of commercial grade basic materials equivalent to the minimum characteristics and performances of the applicable military specification is encouraged.

3.1.1.14 Vibration. Vibration on shafts adjacent to bearings and bearing caps shall not exceed the limits specified in MIL-STD-167 for type II mechanical balance and as specified in 3.1.1.7.4. In addition all equipment shall be designed to conform to the requirements of MIL-STD-167 for type I vibration up to and including 25 c.p.s.

3.1.2 Type II, noncondensing, variable speed operation. The requirements for type II shall be as specified hereinafter and as specified for type I, except 3.1.1.1, 3.1.1.2, 3.1.1.9, 3.1.1.10 to 3.1.1.11.6, inclusive, do not apply.

3.1.2.1 The turbine generator sets shall have the following classification:

Type required	Salient pole.
Steam pressure (at throttle)	As specified (see 6.1).
Steam temperature (at throttle)	Dry saturated (1.0 percent maximum moisture).
Exhaust pressure	50 p.s.i.a.
Rating:	
Kilowatt	As specified (see 3.1.2.7).
Voltage	450 volts a.c.
Class of insulation	Class B or H.
Degree of enclosure	Water-air-cooled.
Cycles	30 through 60 (continuous) 15 to 30 (15 minutes out of each 12 hours).
Type of voltage regulator control	Rotary amplifier or static (Type II or III of MIL-R-2729).
Regulators per switchboard	One main.
Repair parts	As specified herein.
Parallel operation requirements	Momentary parallel operation for transfer of load from one generator to another at any frequency between 30 and 60 c.p.s., and between one generator and the motor generator set at 15 c.p.s.

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3.1.2.2 Each turbine generator set shall be comprised of the turbine, generator, reduction gear, sub-base, and oil coolers mounted as an integral unit. Integral hand lubricating oil pumps for starting shall be included.

3.1.2.3 The turbine generator unit will be used to supply power to drive continuously variable speed pumps operating over a frequency range of 15 to 60 c.p.s. (The actual range required shall be 14.5 to 63 c.p.s., and shall be designated herein as 15 to 60 c.p.s. (see 3.1.2.3.1(c)). The load characteristics shown in table I represent the minimum values required, and shall be used as a guide. The manufacturer shall submit preliminary design information on the turbine generator to the ordering activity for approval prior to release for manufacture. The design information shall include, as a minimum, the guaranteed performance curves of maximum permissible turbine generator power (KW at a stated power factor and rated voltage) versus frequency, and maximum permissible generator current versus frequency for rated voltage.

Table I - Load characteristics for variable speed pump motors.

Running					
Number of pumps	Pump fluid temperature	Frequency	Input KW	P.f.	Volts
4	cold	30	576	0.66	220
3	cold	30	433	.66	220
4	hot	49	1930	.84	359
4	hot	45	1468	.82	330
3	hot	60	2500	.88	440
3	hot	56	2070	.87	411
3	hot	49	1450	.84	359
3	hot	45	1101	.82	330
3	cold	15	96	.28	110
4	cold	15	127	.28	110
3	hot	15	80	.28	110
Starting requirements (per pump).					
Frequency	P.f.	In rush current	Voltage (start)	Voltage (allowable dip)	
60	0.20	5800	440	300	
56	.21	5700	411	275	
49	.23	5600	359	225	
45	.25	5500	330	200	
30	.33	5400	220	110	
15	.57	4800	110	60	

3.1.2.3.1 The design of the unit shall be such as to provide the following:

(a) Voltage directly proportional to the speed over the entire frequency range from 15 to 60 c.p.s.

(b) Supply the normal running loads (see 3.1.2.3) and start pumps under the following conditions:

(1) With 3 pumps running at any frequency between 15 cycles and 49 cycles, with the pump fluid hot, start a fourth pump.

(2) With 2 pumps running at any frequency between 49 cycles and 60 cycles, with the pump fluid hot, start a third pump.

(3) Pumps shall be assumed to reach steady-state speed within 5 seconds. Voltage at the pump terminals shall not dip below the values specified in 3.1.2.3. The time for voltage to return to and remain within the plus or minus 20 percent of the initial voltage shall not exceed 1.5 seconds.

(c) A governor shall control the turbine over a minimum speed range of 14.5 to 63 c.p.s., inclusive, and provide remote speed control over the above stated range. The speed range shall be adjustable over the frequency range specified herein for both manual and automatic operation of the governor. The governor speed changer motor shall be the single speed type without a commutator. The speed change device shall be designed to permit a speed change from 15 to 60 c.p.s. in not more than 3 minutes with a three-pump load on the turbine generator set.

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(d) At constant 100-percent load, minimum rated pressure at the throttle, and speed corresponding to 60 cycles, with an increase of pressure to maximum rated pressure, the speed shall not increase more than 3 percent. In addition, at any constant load and speed at any pressure point throughout the load-speed-pressure range, the periodic fluctuation in speed during rated pressure changes shall not exceed 1 cycle, and upon sudden application of load, the dip shall not exceed 7.5 percent.

(e) A volts per cycle regulator system to regulate automatically the generator output voltage at 7.5 volts/cycle (plus or minus 10 percent) for all frequencies over the range of 15 to 60 c.p.s. The rated voltage for all frequencies shall be maintained by a single adjustment of the system.

(f) Voltage regulator manual control system to permit manual voltage control on a linear volts/cycle basis over the range of 15 to 60 c.p.s. with a single adjustment device only. The voltage tolerances shall be in accordance with the values specified in MIL-R-2729.

(g) Connections to the voltage regulator equipment shall be electrically isolated from the generator power leads by means of isolating transformers or other means approved by the procuring activity.

3.1.2.4 The turbine shall be designed so that no critical speed will be encountered within its operating range or within 10 percent of the upper or lower limits of the operating range.

3.1.2.5 A combined exhaust and relief valve as shown on Drawings 5000-S4824-1385797 and 5000-S4824-1385798 shall be provided by the contractor. This valve shall be sized to limit the pressure inside the turbine casing to 75 p.s.i.a.

3.1.2.6 An overspeed trip and a manual trip shall be provided. The overspeed trip shall be set to operate at 10 percent above the maximum operating speed.

3.1.2.7 The turbine generator sets shall be capable of continuously supplying 2560 KW to meet distribution losses plus pumping power requirements as specified in 3.1.2.3.

3.1.2.8 The required minimum time between operations in the 15-to 30-cycle frequency range shall not be operationally restrictive.

3.1.3 Type III, condensing, constant and variable speed operation. The requirements for type III shall be as specified hereinafter and as specified for types I and II, except that 3.1.1, 3.1.1.1, 3.1.2, 3.1.2.1 and 3.1.2.6 shall not apply.

3.1.3.1 The turbine generator sets shall have the following classification:

Type required	Salient pole.
Steam pressure (at throttle)	As specified (see 6.1).
Steam temperature (at throttle)	Dry saturated (1.0 percent maximum moisture).
Exhaust pressure	As specified (see 6.1).
Rating:	
Kilowatt	As specified (see 3.1.3.1.1).
Voltage	450 volts a.c. (at 60 c.p.s.).
Degree of enclosure	Water-air-cooled.
Cycles	30 through 60 (continuous) 15 to 30 (15 minutes under load) (see 3.1.3.1.2).
Type of voltage regulator control	Rotary amplifier or static (Type II or III of MIL-R-2729).
Regulators per switchboard	one main.
Repair parts	As specified herein.
Parallel operation requirements	Required (see 3.1.3.3).

3.1.3.1.1 The turbine generator sets shall be capable of continuously supplying 2560 KW to meet distribution losses plus pumping power requirements as specified in 3.1.2.3.

3.1.3.1.2 The required minimum time between operations in the 15- to 30-cycle frequency range shall not be operationally restrictive.

3.1.3.2 The turbine generator unit will be used to supply power to variable speed pumps or to supply ship's service power at constant speed 60 cycle operation.

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3.1.3.3 The parallel operation requirements of 3.1.1.9 herein shall apply for constant speed, 60-cycle operation only. In addition, provision shall be made for momentary parallel operation as specified in 3.1.2.1.

3.1.4 Interchangeability. The electrical equipment for type I, type II, and type III generator sets shall be identical. In addition, the maximum possible interchangeability of parts and assemblies of the turbine and gear shall be provided for type I, type II, and type III generator sets.

3.1.5 The generator and exciter rotors shall be balanced to such a degree, that, when assembled with the turbine, the peak to peak amplitude of vibration when measured on the bearing brackets or pedestal shall not exceed 2 mils.

3.1.6 A generator transient reactance not exceeding 0.25 shall be acceptable provided that the maximum voltage dip and recovery time conform to MIL-G-3124 and MIL-R-2729 and the sustained three-phase, short circuit current is at least 3.2 times normal current.

3.2 Equipment manuals. Equipment manuals shall be furnished in the quantity specified (see 6.1); and shall be type I in accordance with MIL-M-15071, except that an introduction shall be included. The following equipment shall be covered in the manual:

Steam turbine and any associated accessories.
Reduction gears.
Lube-oil coolers.
Generator.
Exciter.
Voltage regulator.
Manual control means.
Temperature indicating and alarm devices.
Governor motor.
Special devices on equipment.

Specific instructions for operation shall be stated for the following conditions:

(a) Start-up from cold iron.
(b) Start-up from "standby".
(c) Normal operation.
(d) Securing for standby.
(e) Securing for long periods.
(f) Lay up.
(g) Any other condition the manufacturer may consider a possible operating condition.

3.2.1 Equipment manuals shall contain reduced size prints of all drawings approved for electric equipment and a list of all drawings approved by the Naval Ship Engineering Center (NAVSEC), Electrical Branch, Steam, Gas and Diesel Generators Section indicating the manufacturer's and the NAVSHIPS drawing numbers in parallel columns. Manuals shall be approved by NAVSEC prior to final printing. A copy of the periodic test data on the generator, voltage regulator, and combined turbine generator set shall be included.

3.3 Drawings. All drawings shall be master drawings in accordance with type II or III of MIL-D-1000/2 and shall include information as required by individual equipment specifications.

3.3.1 Installation drawings. Installation drawings shall include the following:

(a) All connections, electrical and mechanical weights, overall dimensions, and other data necessary to clearly describe the equipment.
(b) An outline of the identification plate showing all plate markings and the location of the plate.
(c) Piping diagrams showing recommended connections of oil, water, steam, and exhaust necessary for proper installation and operation of the unit.
(d) Type numbers, ratings (volts, amperes, r.p.m., and kw.) and the name of manufacturer, of all electric equipment.
(e) Information required by paragraph 60 of the Appendix.

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3.3.2 All drawings (unless approved master drawings are on file) shall be submitted to the procuring activity via the Defense Contract Administration Services representative for approval. The installation drawings shall be submitted for approval within 90 days from the date of contract. Drawings for the turbine, gear, lube-oil coolers, generators, exciter, and control equipment shall be submitted for approval within 120 days from the date of the contract.

3.4 Onboard repair parts. Onboard repair parts shall be furnished in accordance with MIL-P-15137.

3.4.1 Standard stock numbers shall be obtained for all parts in accordance with MIL-P-15137.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is reponsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control program. The supplier shall provide and maintain a quality control program acceptable to the Government. The program of quality control shall be in accordance with MIL-Q-9858 and as specified in 4.1.1.1 and 4.1.1.2. The supplier is in no way relieved of the final responsibility to furnish generator sets conforming to the requirements of this specification.

4.1.1.1 Description of procedures. Procedures shall be assembled in manual form, and shall include a flow chart showing the development of the product through the manufacturing cycle. The procedures shall indicate organization and responsibility for control of quality.

4.1.1.2 The measuring and testing device maintenance and calibration schedule shall be available to the Government.

4.2 Type I generator sets. Tests shall be conducted on the type I turbine generator units in accordance with MIL-G-3087. Test results on the first generator and combined turbine generator unit shall be forwarded to NAVSEC, Electrical Branch, Steam, Gas and Diesel Generators Section for approval prior to shipment of the units.

4.3 Types II and III generator sets. Tests shall be conducted on the type II and type III generator units operating at a speed corresponding to 60 c.p.s. in accordance with the periodic test procedures of MIL-G-3087. In addition, sufficient tests shall be conducted at speeds taken over the entire frequency range of 15 to 60 cycles to demonstrate conformance with 3.1.2.3 and 3.1.2.3.1. For these tests, rated KW load shall consist of line losses plus maximum KW load required to be supplied by the unit at any corresponding value of frequency over the entire frequency range of 15 to 60 c.p.s. A test shall also be conducted to demonstrate compliance with the voltage dip requirements of 3.1.2.3.1 for various initial loadings. For the purpose of this test, starting a pump shall be approximately by a balanced three-phase load of 0.7 per unit impedance and 0 to 0.4 lagging power factor. Performance curves as required in 3.1.2.3 shall be taken over the entire frequency range of 15 to 60 c.p.s. with data for plotting the curves taken at maximum frequency intervals of 5 c.p.s. over the 15- to 60-c.p.s. range. Test results shall be forwarded to the procuring activity for approval prior to shipment of units.

4.4 Type II only. The contractor shall demonstrate that with the combined exhaust and relief valve handwheel screwed down and the steam inlet valve fully open, the pressure inside the turbine casing will not exceed 75 p.s.i.a.

4.5 Hydrostatic. All steam fittings and connections which may be subject to boiler pressure, including steam chests, throttle valves, and nozzle chambers shall be tested hydrostatically after assembly to a pressure at least 50 percent greater than the maximum designed boiler safety valve pressure setting.

4.6 The requirements for the turbine overspeed test specified in MIL-T-17523 shall apply except that the maximum reading of the vibration indicator shall not exceed the limits specified in 3.1.1.7.4.

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4.7 Steam consumption shall be measured at 100-percent, 75-percent, 60-percent and 25-percent load with steam minimum rated pressure (1.0 percent maximum moisture). Consumption shall be determined by measuring the condensate either by weighing or using water meters calibrated to the satisfaction of the Defense Contract Administration Service representative.

4.8 Vibration test. Tests in accordance with type I of MIL-STD-167 shall be conducted on all equipments except for the following:

(a) Units which weigh more than 10,000 pounds.
(b) Machinery and equipment units which have been tested up to and including 25 c.p.s. and are satisfactory.

4.9 Test conditions. All of the tests required by 4.2 and 4.3 shall be conducted at minimum rated pressure (1.0 percent maximum moisture). In addition, sufficient tests shall be conducted at various other pressure conditions throughout the pressure range (1.0 percent maximum moisture) to show conformance with performance requirements.

4.9.1 The ambient noise level for the audio noise test need not be guaranteed provided that the test is conducted when the ambient is at a minimum.

4.9.2 A radio noise test on the generator and exciter will not be required.

4.9.3 During the governor tests required by MIL-G-3087, the no-load voltage for sudden load change may be set higher than 105 percent, if necessary, to maintain rated voltage and load after load is applied. In addition, load valve lift data and load speed data may be taken in two separate runs.

4.10 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, (see 6.3.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts (see 5.3).

5.1.1 Generator set.

5.1.1.1 Preservation and packaging. Preservation and packaging, which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation, and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. Mechanical and electrical repair parts shall be preserved and packaged level A, packed level C, and marked respectively, in accordance with MIL-E-16298, MIL-P-17286, and MIL-E-17555, as applicable (see 5.3).

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1):

(5.2.1 The following provide various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

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5.2.1.1 Generator sets. The equipment, accessories, technical publications, and repair parts shall be preserved and packaged by level A or C; packed by levels A, B, or C; and marked, as specified, in accordance with MIL-E-16298, MIL-P-17286, and MIL-E-17555 (see 5.3).

5.2.1.2 Onboard repair parts stowage boxes. The level of cleaning, preservation, packaging, and packing, and the requirements for onboard repair parts stowage boxes shall be as specified (see 6.1).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION:

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Steam pressure (at throttle) (see 3.1.1.1, 3.1.2.1, and 3.1.3.1).
(d) Exhaust pressure (see 3.1.1.1, 3.1.1.10, 3.1.2.1, and 3.1.3.1).
(e) Specify when weight and size restrictions are required (see 3.1.1.3).
(f) Air ejector design pressure (see 3.1.1.11.1).
(g) Quantity of equipment manuals to be furnished (see 3.2).
(h) Level of preservation, packaging, packing, and marking if other than as specified in 5.1 (see 5.2).
(i) Requirements for onboard repair parts stowage boxes (see 5.2.1.2).
(j) When polystyrene (loose-fill) material is approved (see 5.3).

6.2 Shock tests (see Appendix). The following should be considered in connection with the shock tests:

(a) Preservation, packaging, and shipment of equipment to and from the shock testing activity.
(b) Shock tests and any repair or redesign resulting from the shock tests.
(c) Post-shock test teardown, examination, repair, modification and testing of the shock tested equipment at the contractor's plant.
(d) Furnishing of contractor's representatives to supervise installation of equipment and initial checkout at the shock testing activity (see Appendix).
(e) Furnishing the shock testing activity drawings of the equipment as necessary to provide complete installation data, including weights and centers of gravity of the set and detached accessories.
(f) Furnishing to the test activity all required information regarding the method of operating the equipment during the test and what capacities of electric power, cooling water, lube oil, air and so forth, will be required.
(g) Furnishing of information to the test activity as to when equipment will be delivered for test. In addition, NAVSEC, Steam, Gas and Diesel Generators Section should be notified at least 45 days in advance of scheduled tests so that arrangements can be made to witness the test, if desired. The contractor shall submit, along with the notification, an equipment data sheet covering the equipment to be tested and containing the information shown in the Appendix.

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6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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APPENDIX

REQUIREMENTS FOR HIGH IMPACT (HI) SHOCK EQUIPMENT

10. SCOPE

10.1 This appendix covers the design, test, and acceptance requirements for turbine generator sets which must withstand HI shock.

20. APPLICABLE DOCUMENTS

20.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the appendix to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-857 - Bolts, Nuts, and Studs.
MIL-S-901 - Shock Tests, HI (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-21952 - Steel (HY-80 and HY-100) Bars, Alloy, Hot Rolled.
MIL-S-22664 - Steel, Alloy, Extrusion, Structural Shapes (HY-80 and HY-100).
MIL-S-22958 - Steel, Alloy, Shapes, Hot Rolled Structural (HY-80 and HY-100).
MIL-S-23008 - Steel Castings, Alloy, High Yield Strength (HY-80 and HY-100).
MIL-S-23009 - Steel Forgings, Alloy, High Yield Strength (HY-80 and HY-100).

STANDARDS

MILITARY

MS17828 - Nuts, Self Locking, Hexagon, Regular Height, (Nonmetallic Insert) 250°F., Nickel-Copper Alloy.
MS17830 - Nuts, Self Locking, Hexagon - Regular Height, 250°F. (Non-Metallic Insert) 300 Series Corrosion Resistant Steel.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 The requirements of MIL-S-901 shall apply except as otherwise specified herein.

30.2 Turbine generator sets and all attached and detached accessories shall conform to the requirements for Type A, grade A, class 1 of MIL-S-901 and the shock test acceptability criteria specified in 30.4. Acceptance of generator set and accessory designs which have been subjected to shock tests shall be based on the ability of the equipment to conform to acceptability criteria, as revealed by its performance during and after shock tests and the results of post-test examination and operating tests conducted at the supplier's plant. In the event that the acceptance criteria are not met, acceptance will also depend upon the corrective action proposed by the contractor. A detailed report of the inspection along with the contractor's proposals for shock hardening (if necessary) shall be forwarded to NAVSEC Electrical Branch, Steam, Gas and Diesel Generators Section. Approval by NAVSEC of the results of shock tests, the pre-shock and post shock test inspections, and the corrective action shall constitute acceptance of the design.

30.3 Intended function. The intended function of shockproof generator sets is to continuously deliver electrical power conforming to required performance characteristics throughout and after being subjected to shock of the magnitude experienced during shock tests as specified in MIL-S-901. Further, a shockproof generator set, when subjected to shock, shall not suffer damage to the extent that it creates a possible hazardous situation such as fire or injury to operating personnel, or such as to result in likely failure of major units of the set if the set is not immediately shut down for corrective action. In determining if equipment has passed shock tests, the criteria in 30.4 herein shall be used.

30.4 Acceptability criteria for shock tested equipment. In determining if a shock tested equipment has passed shock tests, the criteria specified hereinafter shall apply. NAVSEC shall be the final judge of whether an equipment shall be accepted or rejected on the basis of shock test requirements; however, the manufacturer shall have the option of convening a conference with NAVSEC to discuss questionable situations involving acceptance or rejection of shock-tested equipment.

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30.4.1 Damage as characterized or as specified herein shall not be permitted as a result of shock tests. In the following, as elsewhere in this section, shock tests are sometimes equated to an attack upon a Navy ship.

(a) Damage which causes or necessitates reduction of equipment output to a level below rated for any length of time within 30 days after attack.
(b) Damage that renders inoperative any item or device installed to protect the equipment or operating personnel, such as speed limiting governors, overspeed trips, high pressure relief valves, and low lube oil pressure alarms.
(c) Shock-induced actuation of safety devices specified in 30.4.1, item (b).
(d) Damage or shock-induced misadjustment of speed governing systems that results in inability to maintain speed automatically within the limits imposed by the individual equipment specifications until repairs or adjustments are made.
(e) Loosening or shifting of any item affixed to a rotating part.
(f) Damage that causes any assembly or subassembly to come adrift from any of its intended restraints.
(g) Damage that requires repairs of any type to be made before equipment operating parameters can be adjusted to suit operational requirements.
(h) Damage that creates a safety hazard to personnel or equipment.
(i) Complete fracture of any unit or welded assembly.
(j) Complete stripping or complete failure by gross distortion of all engaged threads (male or female) of any threaded part.
(k) Increase of dynamic unbalance of any rotating part or assembly to such an extent that the vendor must re-balance the affected item prior to delivery of the equipment to NAVSEC or prior to conducting field balancing operations.

30.4.2 Damage from shock tests as characterized or specified herein will be permitted to the extent or under the conditions listed for each item.

30.4.2.1 Damage that shortens the useful life of wearing parts by initiating or aggravating some long-term detrimental effect such as wear, fatigue, or erosion is permissible if the useful life of wearing parts is reduced by no more than 25 percent.1/

30.4.2.2 Damage resulting in minor leaks of fluids or gases is permissible insofar as no portion of this appendix is violated by any consequences of the leakage.

30.4.2.3 Loosening of hold-down is permissible if no portion of this appendix is violated as a consequence of such loosening.

30.4.2.4 Damage that brings about a requirement to adjust control apparatus or equipment setpoints in order to regain the pre-attack mode of operation, or damage that brings about a requirement for timely repairs or compensatory action in order to prevent further (and eventually unacceptable) damage that is related to increased wear rates or other long term effects, is permissible only if the following conditions are met:

(a) If NAVSEC agrees that the damage or its effects are of such a nature that timely and proper damage diagnosis and compensation would certainly be made by ship's force personnel familiar with the equipment. (For purposes of this appendix, it is considered that misalignment (if any) or shafting cannot be corrected by ship's force).
(b) If the equipment status before adjustment/repairs are made is such that no portion of this appendix is violated by the immediate effects of the damage or by requirements for repair of the damage.

1/ Due to difficulties associated with estimating reduction of useful life, equipment acceptability shall be determined with consideration for the accuracy of the estimate. In cases where it is especially difficult to estimate percent reduction of useful life, equipment will not be rejected for this cause unless in the opinion of NAVSEC that the limits specified herein have been grossly exceed. In cases where the availability of test or maintenance information permits a more accurate estimate of useful life reduction to be made, appropriately less leniency will be allowed when determining acceptability of the equipment on this basis.

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30.4.2.5 Damage that results in continuous or intermittent rubbing contact between stationary and moving parts (during equipment operation) in areas where it was not originally intended that such contact be made is generally not permitted. NAVSEC may choose to permit on a case basis, rubs which quickly wear away without causing adverse effects that otherwise violate any portion of this appendix.

30.4.2.6 Damage that brings about (until repairs are made) a requirement for slight to moderate increase of manual effort, maintenance, or attention on the part of equipment operators in order to maintain a satisfactory operating mode or a suitable machinery space environment may be permissible, when specifically approved by NAVSEC, Electrical Branch Steam, Gas, Diesel and Generator Section.

30.5 Design

30.5.1 The generator set subbase hold down bolt holes provided for securing equipment to the ship's foundation shall be designed for use with nonfitted bolts conforming to grade 2, types I, II, and III of MIL-B-857 and self-locking nuts in accordance with MS17828 and MS17830. Maximum clearance between bolt holes and bolts shall not exceed the following:

Nominal bolt diameter	Maximum diameter of hole
3/4 inch or smaller	Nominal bolt diameter plus 1/32 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

30.5.2 When carbon steel castings, forgings, or plates are used in bedplates, baseplates, feet, stanchions and subbases, and the thickness is 1/2 inch or over, the following shall apply:

(a) Stresses of 10,000 p.s.i. and under steel may be used as cast, forged, or rolled.
(b) Stresses in excess of 10,000 p.s.i., steel castings or plates shall be normalized by heating to 1650° ± 25°F., holding a minimum of 30 minutes per inch of full thickness, followed by cooling individually from the holding temperature in still air.
(c) Steels for normalizing shall be fully killed and fine grained.
(d) All test specimens shall be selected after final heat treatment.
(e) All normalizing shall be accomplished prior to welding.

30.5.3 When high yield strength materials (materials with yield strength of 75,000 p.s.i. or more) are used, by reason of choice or requirement, to meet design stress levels, or to satisfy resistance to shock, or both, HY-80 or HY-100 in accordance with MIL-S-21952, MIL-S-22664, MIL-S-22958, MIL-S-23008 or MIL-S-23009, as applicable, shall be used.

40. QUALITY ASSURANCE PROVISIONS

40.1 General inspection. Complete generator sets and detached accessories shall be subjected to and successfully pass shock tests.

40.2 Test requirements.

40.2.1 All shock tests shall be conducted in accordance with the grade A machinery requirements and type A tests of MIL-S-901 and as follows:

(a) The generator set package (prime mover, gear, generator, exciter, subbase, heat exchanger(s) and all attached accessories) any detached accessories such as condenser, air ejector, and accessories needed to control and operate the set shall be shock tested as a principal unit if the weight size and arrangement limitations are not exceeded. Detached accessories not tested with the generator set package shall be individually tested using the applicable shock testing machine referenced in MIL-S-901 and in accordance with the individual equipment specification (For example, MIL-R-2729 - Regulator Sets, Voltage).
(b) Assemblies or sub-assemblies comprising the equipment to be shock tested shall be those actually made for and furnished under the present contract.

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(c) All assemblies and subassemblies comprising the equipment furnished for shock tests shall comply with individual equipment specification requirements. All required performance tests shall have been satisfactorily completed prior to shock testing.[2/]

(d) The manufacturer shall carefully inspect equipment which will be shock tested in order to detect any flaws or undesirable conditions that might contribute to shock test damage (see[3/]), or be mistaken for shock damage during the post-test inspection.

(e) All running clearances, alignments, runouts, and mechanism set points[4/] that are normally established within the equipment and its ancillary devices by the contractor shall be properly established in the equipment before shock tests, even though it may be the contractor's usual practice to establish certain of these settings after installation of the equipment aboard ship. If the contractor desires, setpoints may be established by setting or adjusting the device in question in such a manner that the correct setpoints are theoretically achieved (that is achieved without conducting actual tests). If setpoints are obtained in this manner, however the contractor shall document the pre-test status of all adjustments, dimensions, and characteristics that could possibly cause change of setpoints as a result of shock test damage. If the contractor desires that certain alignments, clearances, or setpoints be established by the shock testing facility as the equipment is erected for shock tests (necessary, for instance, if equipment is disassembled for shipment), the cognizant shock testing facility shall be so informed at least 2 months before the equipment is to be delivered to the shock testing facility. Due to facility limitations, the contractor may be required to provide personnel or equipment, or both necessary to accomplish such work.

40.3 Equipment mounting during test.

40.3.1 Heavy weight equipment. Shock tests shall be conducted on the floating shock test platform specified in MIL-S-901.

40.3.1.1 The generator set package shall be mounted upon a foundation whose dynamic characteristics simulate those of the intended shipboard foundation as closely as possible. In no case shall the foundation used for shock testing purposes possess less stiffness (or have a lower natural frequency) in any direction than the stiffest intended shipboard foundation for the item tested. Accessories, if tested with the generator set package shall be mounted on a fixture designed to accommodate the accessory and shall be welded to the inner deck. Bulkhead mounted accessories shall be mounted on a fixture designed to accommodate the accessory and shall be welded to the side of the barge.

40.4 Test procedure. Shock tests shall be conducted in accordance with MIL-S-901 and as specified hereinafter.

[2/] Exceptions: If field balancing techniques must be employed to meet contract noise and vibration requirements, field balancing shall be accomplished by the contractor after the completion of shock tests. Request for other exceptions to 40.2.1 item (c) requirements will be considered on a case basis by NAVSEC.

[3/] For purposes of this appendix "damage" is defined as any permanent change of the status or condition of any equipment that results from shock testing.

[4/] "Setpoints" define the input-output relationships of any mechanisms or device associated with the equipment to be shock tested. For instance, the setpoint or an overspeed tripping device could be "actuation (output) at 10,700 revolutions per minute (r.p.m.) (input)."

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40.4.1 Equipment shall be rotated by air pressure during each blow or shot required by MIL-S-901 at a speed as close to rated speed as practical with the available air supply facilities[5/]. Assemblies and accessories, attached or detached, which contain a fluid during normal operation (such as generator air coolers, condensers, air ejectors, fresh water coolers and so forth) shall contain a fluid of comparable weight during all blows. The equipment shall be examined for damage after each blow. All mounting bolts of the equipment and shock machine mounting shall be tightened before each blow. Correction of damage during the progress of tests will be permitted, and the test may proceed to the next level of shock after the damage is corrected.

40.4.2 Equipment test status will generally be selected in such a manner that the most adverse situations for shock (within the realm of realistic or normal operation of the equipment) are represented during tests. For example, if a valve stem is most likely to fail under shock when the valve is half open, it would probably be tested in the half-open position. The contractor shall design the equipment to pass shock tests regardless of the operating mode and equipment status selected by NAVSEC for tests.

40.4.3 Retesting policies. Damage violating acceptance criteria which occurs during the most severe shot or blow of a shock test series shall be cause for suspension of tests until such time as the contractor's proposed corrective action is approved by NAVSEC, Electrical Branch, Steam, Gas and Diesel Generators Section. Retesting of the equipment at the most severe test level shall be required after the approved corrective action has been completed unless NAVSEC and the equipment supplier agree that the suitability of modifications required to make an equipment acceptable from the standpoint of shock requirements can unquestionably be demonstrated by calculations or practical consideration. If a heavy weight equipment is rejected solely on the basis of shock test damage to an item that can be readily detached from the principal equipment, and if this item can be realistically shock tested on a light or medium weight shock machine, NAVSEC may choose to permit testing of modifications made to the affected item in this manner.

40.4.4 Instrumentation and measurements during the tests shall be as follows:

(a) Standard instrumentation for heavy weight shock tests shall consist of one or more velocity transducers mounted upon the floating shock platform deck, to permit documentation of shock inputs. The shock facility shall determine the number and location for transducers to be used for this purpose. If the contractor desires that additional instrumentation[6/] be used to document equipment motion or other shock induced equipment reactions he shall submit proposed requirements to the cognizant shock facility and to NAVSEC as far in advance of tests as possible. The use (or extent of use) of additional instrumentation is subject to limitations imposed by facility requirements and shock testing schedules.
(b) Initial start-up and checkout shall be conducted under guidance of the manufacturer's representative.
(c) Scribe lines or punch marks shall be used to indicate exact location of all major attached accessories and assemblies (such as blowers, pumps, turbine and generator feet, and covers and so forth) prior to shock tests so that any relative motion during shock blows can be measured and recorded.
(d) The watersides of all coolers and condensers shall be hydrostatically tested before and after the test series.
(e) Air-gap measurements shall be taken before start of tests and after each blow or shot.

[5/] To prevent icing in the turbine steam path, air should not be supplied to turbines at pressures higher than 100 pounds per square inch gage (p.s.i.g.) and operating time (using compressed air) should not exceed 10 minutes per 1 hour period.

[6/] Transducers (with appropriate readout devices) for remote measurement of velocity, acceleration, vibration, strain, r.p.m., voltage, and pressure, are generally available at shock-testing facilities. High speed motion picture cameras are usually available. The number of information channels may be limited to ten or less at some shock facilities.

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(f) Insulation resistance and dielectric strength shall be measured prior to start of testing and after the final blow or shot.
(g) A thorough visual examination shall be made after each blow or shot and any indications of movement or distortion recorded.
(h) Generator cooler covers shall be removed after each blow or shot for visual examination of cooler and any excitation parts.
(i) Turbine nozzle clearance (minimum) and turbine thrust bearing clearance shall be measured after each blow or shot.
(j) All turbine throttle valve tripping devices shall be activated before each shot or blow, and the throttle valve opened. After each test, it shall be verified that shock-induced tripout did not occur and that all tripping devices operate properly.
(k) For tests on heavyweight equipment, provision shall be made for remote observation of lubricating oil pressure levels to bearings. Provision shall also be made for remote shutdown of the equipment.
(l) Gear tooth contact (percent), runout of principal rotating parts, and alignment of principal shafting shall be determined before and after the test series.

40.5 Post-shock test inspection. All shock tested equipment shall be subjected to a post-shock test inspection, to be conducted by the contractor. Basically, this inspection shall consist of disassembly and examination of the equipment, followed by operating tests of the shock-tested unit. Post-shock inspection shall be as specified hereinafter.

40.5.1 Representatives of NAVSEC shall have the priviledge of witnessing any or all post-test examination and operating tests. The contractor shall advise NAVSEC of the anticipated post-shock test inspection schedule, so that arrangements may be made for NAVSEC representation. NAVSEC or the contractor shall have the option of convening a conference to discuss requirements for the post-test inspection. The contractor's schedule for post-test activities shall not be affected by these requirements.

40.5.2 Prior to conducting operating tests, the shock-tested equipment shall be disassembled sufficiently to permit the inspections, specified in 40.5.2.1 through 40.5.2.9, as applicable, to be made. The contractor's usual methods for conducting the applicable inspection may be followed.

40.5.2.1 All wearing and moving parts or assemblies shall be visually examined for any sign of damage or excessive wear.

40.5.2.2 Shafting alignment and close-running clearances between rotating and non-rotating parts shall be checked.

40.5.2.3 Clear backlash and gear tooth contact shall be checked.

40.5.2.4 All moving or loose-fitting parts shall be checked for freedom and smoothness of movement, or absence of excessive play or looseness or both.

40.5.2.5 All bolted connections shall be examined and all bolts shall be checked for any sign of looseness.

40.5.2.6 All dowelling shall be checked for looseness. Dowels and dowel holes shall be checked for deformation.

40.5.2.7 All welds shall be examined visually. Welds in the vicinity of damage structure and welds suspected of being highly stressed shall be checked with dye-penetrant or other appropriate methods to detect small cracks that are not readily visible.

40.5.2.8 Assemblies or parts of any type that are suspected of being highly stressed or damaged in any manner as a result of shock tests shall be visually examined. Appropriate measurements shall be made as necessary in such cases to reveal damage.

40.5.2.9 Dynamic balance tests shall be performed upon turbine rotors, and upon any rotating parts or assemblies suspected by NAVSEC of having sustained shock test damage that could cause a significant increase of dynamic unbalance. If the applicable shafting is solidly coupled to other shafting elements, dynamic balance tests shall be conducted with the shafting coupled if the manufacturer originally balanced the shafting in this manner 7/.

7/ If solidly-coupled shafting for which dynamic balance tests are required must be disassembled for shipping purposes, runout measurements shall be made before disassembly of the shafting.

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40.5.3 Assemblies with press-fitted parts, sealed gear boxes on small motors, wound motor rotors, and mechanisms with set points established during original assembly are examples of items which do not lend themselves well to complete disassembly for examination. If performance of such items during shock tests and external examination do nor reveal any sign of damage, and if conclusive tests can be performed to establish that the item functions properly in accordance with the applicable equipment, NAVSEC will generally waive requirements for complete disassembly.

40.5.4 Requirements for operating tests of shock-tested equipment shall be as specified hereinafter. NAVSEC shall require that misalignment or other equipment degradation resulting from shock be reestablished for operating tests if there is doubt or disagreement (between NAVSEC and the contractor) regarding the effect of such damage upon operation.

40.5.4.1 Operating tests of all shock-tested generator sets shall be as specified in 40.5.4.1.1 through 40.5.4.1.6.

40.5.4.1.1 One run shall be made at no load (or at the lowest load possible) in which speed is gradually increased to rated or full speed, maintained at full speed until thermal conditions have stabilized, and then increased gradually until rated overspeed is reached. During this run, special attention shall be given to detecting any unusual or unwarranted noise or vibration emitted by the equipment.

40.5.4.1.2 One run shall be made at rated load and rated speed for a minimum of 4 hours, with turbine and generator performance monitored during the run.

40.5.4.1.3 Speed limiting governors shall be tested three times with the equipment running at no load (or at the lowest load possible) by approaching overspeed and verifying that they are working properly at their intended set points. The throttle valve shall be wide open at the conclusion of each trial, and steam pressure and temperature at the throttle shall be at the rated maximum operating levels.

40.5.4.1.4 All safety devices such as overspeed trips, high back pressure trips, and relief valves shall be tested in an appropriate manner three times to determine if they are operating properly at the correct setpoint(s).

40.5.4.1.5 All ancillary devices such as pumps, valves, motors, speed governors, sealing steam regulators, and gages shall be operationally tested to verify that they have not sustained damage that affects operation or setpoint(s). Generally, operating tests of ancillary devices can be accomplished during operating tests of the primary equipment; separate bench tests of each ancillary device are not necessarily required.

40.5.4.1.6 In general, it is required that all routine tests of MIL-G-3087 and the applicable individual equipment specifications be performed. It shall be demonstrated that equipment performance satisfies the requirements of these specifications.

40.5.5 Applicable post-test inspections shall be conducted at the shock testing facility (instead of at the contractor's plant) under the circumstances specified in 40.5.5.1 through 40.5.5.3.

40.5.5.1 If the equipment must be partially disassembled for shipping purposes, measurements shall be taken as necessary to document the status of the equipment before disassembly.

40.5.5.2 All measurements identified with items sensitive to shipping damage shall be taken before shipment.

40.5.5.3 In cases where there is any specific evidence indicating that the equipment may be rejected on the basis of post-test inspection results, measurements or inspections applicable to the suspected damage shall be conducted before the equipment is shipped, shock facility schedules and facilities permitting.

40.6 Examination to follow operating tests. Upon the satisfactory completion of operating tests, the equipment supplier shall examine the equipment to ascertain the following:

(a) Condition and clearance of all journal and thrust bearings associated with principal shafting.
(b) Gear tooth contact (percent) and condition (visual) of gear tooth.

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(c) Turbine nozzle clearance (minimum) and total float of turbine rotor (with all turbine thrust bearing shoes removed).
(d) Condition of generator brushes and slip rings.

40.7 Inspection reports.

40.7.1 Pre-shock inspection report. The contractor shall document the pre-test status of all running clearances, alignments, runouts, and setpoints that were checked or established in the process of complying with 40.2.1 item (e). Also, the contractor shall take measurements or run tests as necessary to establish reference data for all post-test inspections (except full scale operating tests) that he plans to conduct at his plant (see 40.5.5). This information shall be forwarded to NAVSEC and to the cognizant shock facility at least 4 weeks in advance of delivery of the equipment for shock testing. Methods used in taking measurements shall accompany all data. If any portion of the pre-test inspection information furnished by the contractor is incomplete, unclear, or otherwise unsuitable, the shock facility shall take or repeat, under the technical supervision of the contractor's representative, the applicable pre-test measurements if the contractor cannot supply the data required.

40.7.2 Post-shock test inspection report. A detailed letter report of all post-shock test examinations operating tests (including any inspections performed at the shock facility) shall be prepared by the equipment contractor. Five copies of this report shall be submitted to NAVSEC, Electrical Branch, Steam, Gas, Diesel and Generator Section for approval. The report shall include:

(a) The results of the inspection in detail.
(b) Photographs to show damaged parts and evidence of significant relative movement or impact of parts.
(c) The contractor's recommendations for design improvements to "shock harden" the equipments.

40.8 Disposition of shock tested equipment.

40.8.1 Unless otherwise specified in invidual equipment specifications, shock-tested equipment will be acceptable for delivery under a contract or order provided the following requirements are met:

(a) All damage resulting from shock tests is corrected. The method of correction shall be approved by NAVSEC, Electrical Branch, Steam, Gas, Diesel and Generator Section.
(b) All bearings shall be replaced where inspection reveals that bearing clearances exceed the maximum clearance allowed by manufacturing tolerances. Where major damage to bearings is discovered, appropriate redesign corrective action shall be taken.
(c) Any design improvements necessary to "shock harden" are incorporated in the unit.

50. ARRANGEMENTS FOR SHOCK TESTING

50.1 Light weight and medium weight tests of accessory items (if used). Shock tests may be made at any commercial or Government activity having an approved test machine. The supplier shall make arrangements for such tests direct with the activity of his choice. A list of such activities having approved shock test machines in the area of the supplier's plant may be obtained by written request to NAVSEC, Electrical Branch, Steam, Gas and Diesel Generators Section.

60. MARKING

60.1 Equipment which conforms to the requirements specified herein shall include the following marking on the combined-generator set installation drawing and on the detached accessory assembly drawings:

(a) Class of shockproofness.
(b) Reference to shock test reports and approval letter or shock design calculations and approval letter.
(c) Type of mounting fixture used.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated in reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1426 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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